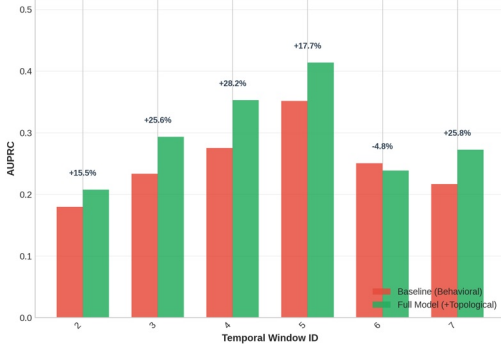
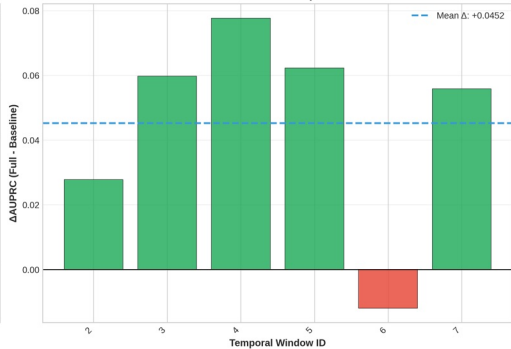


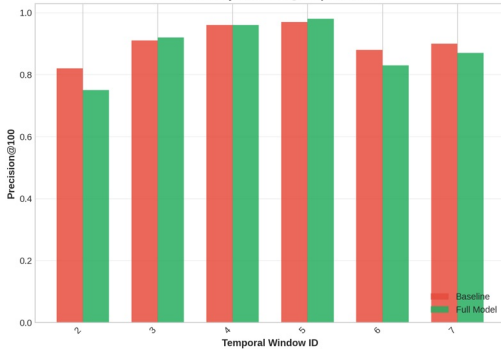
Ablation Study: AUPRC per Temporal Window



Per-Window AUPRC Improvement



Ablation Study: Precision@100 per Window



ABLATION STUDY: STATISTICAL SUMMARY

AUPRC (Area Under Precision-Recall Curve)

Mean Δ: +0.0452 ± 0.0324
t-statistic: -3.4241
p-value: 1.8755e-02
Significant: ✓ YES (α=0.05)

ROC-AUC

Mean Δ: +0.0300 ± 0.0148
p-value: 4.2431e-03

Precision@100

Mean Δ: -0.0217 ± 0.0337
p-value: 1.7626e-01

CONCLUSION:

Topological features provide STATISTICALLY SIGNIFICANT improvement in fraud detection performance.